

BINF6999 Research Report — Grading Rubric

Project: A Mechanism-Informed Gene Signature for Regenerative vs. Fibrotic Tendon Healing

Student: Fangyi Li

Supervisor: Dr. Wilder Scott

Bioinformatics Report: Bundle Specifications

Bundle C (70–80): The Foundational Level

Minimum requirements to demonstrate basic competency.

- **Abstract** is not necessary.
- **Introduction:** Provides sufficient set-up and literature review that leads to a knowledge gap and the biological question/hypothesis (the TSPC vs. T-FAP fate decision and its role in tendon healing). Explains how bioinformatics tools will address this question, and provides a basic explanation of its importance and novelty. The two-arm project structure (mechanistic arm and applied arm) should be introduced.
- **Methods:**
 - Identifies the final datasets used, including source, size, and key characteristics (GSE244921 — Cherief 2023 scRNA-seq; PRJNA506218 — Harvey 2019 scRNA-seq).
 - Lists the primary programming language (Python) and appropriate libraries/packages/pipelines (e.g., Scanpy, pySCENIC, CellRank, LIANA).
 - Provides a high-level description of the analyses performed (bioinformatics, statistics, biological interpretation).
- **Results:** Includes at least two primary deliverables (e.g., a data visualization and a summary table) that directly relate to the biological question/hypothesis.
- **Discussion:**
 - Summarizes the main findings of the research.
 - Discusses their importance in the field of tendon biology and regenerative medicine.
- **Formatting:** The report is approximately 15 pages (excluding references, appendices, figures, and tables), is professionally formatted, and includes a complete list of references in [reference style determined by advisor].

Bundle B (80–90): The Practitioner Level

Includes all requirements from Bundle C, plus the following:

- **Hypothesis Integration:** Evaluates and integrates the multiple objectives and pipelines of the project (e.g., TF network inference, fate trajectory analysis, cell-cell communication) toward a coherent mechanistic narrative.
- **Technical Rigor:**
 - Includes a detailed Materials and Methods section describing data cleaning, normalization, and any pre- and post-processing steps taken (e.g., QC filtering criteria, normalization approach, HPC pipeline execution).
 - Specifies and justifies the statistical tests or algorithms used to validate the results (e.g., Wilcoxon tests for differential regulon activity, rank aggregation for ligand-receptor scoring).
 - Figures and tables can stand on their own, with clear legends, labels, and visual elements.
- **Discussion:** Provides a comprehensive interpretation of the results, both from a bioinformatics and biology perspective, explaining the "so what?" factor.

Bundle A (90+): The Researcher Level

Includes all requirements from Bundles C and B, plus the following:

- **Novelty and complexity:** Of the biological question, bioinformatics approach, pipeline development, statistical analyses, narrative of the report, quality of the data created, quality of the figures and tables.
- **Reproducibility:** Includes a section on computational reproducibility, specifying how environments are managed (e.g., Conda environments, SLURM scripts on HPC, GitHub repository structure).
- **Advanced Analysis:**
 - Weaknesses and scope of the datasets and/or data collection procedures are described (e.g., cross-dataset integration limitations, lack of RNA velocity data, pseudotime caveats).
 - Assumptions behind methodologies are described and defended.

- **Critical Evaluation:** Provides a robust discussion of potential pitfalls, technical limitations, how biological noise might impact the findings, biological implications are discussed in depth, and realistic ideas for future studies.
 - **Professionalism:** Writing is of "submission quality," requiring zero to minimal editing, with perfectly formatted citations and high-resolution figures (if any conceptual diagrams are included).
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The "Unsatisfactory / Fail" Category

A grade of **F** (or Unsatisfactory) is assigned if the report meets any of the following conditions:

- **Incomplete:** Fails to meet the basic requirements of Bundle C.
- **Plagiarism:** Contains unoriginal work or fails to cite data sources and literature.
- **Formatting breach:** Significantly misses the 15-page length requirement (either too short to cover the material or excessively long).